

Weekly Roadmap Week 7: Clustering

How to use this roadmap: This is not a schedule, it is a sequence. Do not skip ahead. Do not read resources before attempting. The struggle is the point. Each day builds on the previous one. If you finish early, go deeper and do not move to the next day's task.

Before You Start Mindset Check

Read this once before Day 1:

You are going to receive a dataset with no labels, no instructions on which algorithm to use, and no guarantee that your first attempt will make sense. That is intentional.

Your job this week is not to produce perfect clusters. Your job is to make decisions, observe consequences, document failures, and arrive at Sunday's session with a story about what you tried and what you learned.

The session will not teach you what you could have read. It will dissect what you experienced.

Day 1 : First Encounter

Time estimate: 2–3 hours Goal: Get your hands on the data. Build the customer matrix. Reach a Minimum Viable Solution (MVS) by end of day.

What to Understand First: Before touching the dataset, make sure you can answer these in your own words:

- What is a feature matrix and why does clustering require one row per entity?
- What are Recency, Frequency, and Monetary what does each measure about customer behavior?
- What does K-Means do at a high level what is a centroid, and what does the algorithm update at each step?
- What is the difference between a raw transaction record and a customer-level aggregation?

What to do:

Download the UCI Online Retail II dataset. Load it. Do not clean anything yet just look at it. Print the shape, the dtypes, the first 10 rows, and the missing value counts. Write down in your own words what one row represents and what problems you already see.

Then clean the data, For every step, write a comment explaining why not what. "Removed rows where Quantity < 0" is what. "Removed because negative quantity implies a return or data error including it would inflate Monetary values" is why.

Build your RFM feature matrix. Start with just three columns: Recency, Frequency, Monetary. One row per customer. If your matrix has more rows than unique CustomerIDs, stop and find the bug.

Run K-Means on your RFM matrix with k=3. Do not tune anything yet. Just get it running. Plot a scatter plot of Frequency vs Monetary coloured by cluster. Look at it.

Your MVS checkpoint:

- Raw data loaded and cleaned ✓
- Customer RFM matrix built ✓
- K-Means running and producing labels ✓
- One scatter plot showing clusters ✓

Do not move on until you hit this checkpoint.

Failure Log entry write this tonight: What did you expect the clusters to look like before you ran the algorithm? What did you actually see? Were they the same?

Day 2 : Go Deeper on K-Means

Time estimate: 2–3 hours Goal: Understand what you built yesterday. Make deliberate decisions about k and initialization.

What to Understand First: Before tuning anything, make sure you understand:

- What is inertia (within-cluster sum of squares) what does it measure and what does minimizing it mean?
- What is the Elbow Method what are you plotting and what does the "elbow" represent?
- What is the Silhouette Score what do a(i) and b(i) represent geometrically?
- Why does the starting position of centroids affect the final result what does this tell you about the K-Means objective function?
- What is K-Means++ initialization how does it differ from random initialization in its selection strategy?
- Why does feature scaling matter for distance-based algorithms what happens to a feature with a large range versus a small one?

What to do:

Before touching code look at your Day 1 clusters. Can you describe each cluster in one sentence? If you cannot, your features or your k may not be right yet.

Now run the Elbow Method and Silhouette Score across k=2 to k=10. Plot both. Do they agree on the best k? If they disagree, do not just pick the one with the better number write down why they disagree and which one you trust more for this specific business problem.

Run K-Means five times with `init='random'` and five times with `init='k-means++'`. Record the inertia for each run. Compare the standard deviation. What does instability in random initialization tell you about the objective function?

Scale your features if you have not already. Then re-run K-Means and compare results to your unscaled version. Did your clusters change? Why?

Questions to answer in your notebook today:

- What k did you choose and why?
- Did scaling change your clusters? What does that tell you about distance metrics?
- Which initialization method would you use in production and why?

Day 3 : Hierarchical Clustering + Mid-Week Hints Released

Time estimate: 2–3 hours Goal: Build the dendrogram, choose a cut point, compare to K-Means.

What to Understand First: Before building the dendrogram, make sure you understand:

- What is hierarchical clustering? What is the difference between Agglomerative (bottom-up) and DIANA/Divisive (top-down)?
- What is a dendrogram, what does the height of a merge represent, and what does cutting it at a certain height produce?
- What is a linkage method? What is the difference between Ward, Complete, Average, and Single linkage in terms of what distance they measure between clusters?
- What is BIRCH and why does it exist? What problem does it solve that standard Agglomerative clustering cannot handle?
- When would you choose hierarchical clustering over K-Means what does the tree structure give you that a flat partition does not?

What to do:

Start with a sample of your data (200–300 customers) and compute the linkage matrix using Ward method. Plot the dendrogram. Stare at it for a few minutes before doing anything else. Where would you cut it? Write down your reasoning before you compute anything.

Now cut it. Run AgglomerativeClustering with the number of clusters your dendrogram suggested. Profile the clusters compute mean RFM per cluster. Do they tell the same story as your K-Means clusters?

Run a second linkage method (complete or average). Compare the cluster assignments. Did the linkage method change who ended up together? Why?

Mid-week hints are released today. Read them only after you have attempted the section you are stuck on, not before.

Check the forum today. Read at least two posts from peers. You do not have to have the answer but respond with one clarifying question that helps them isolate where they are stuck.

Questions to answer in your notebook today:

- How did you decide where to cut the dendrogram? Was it math, visual judgment, or both?
- Did Ward and your alternative linkage method agree on cluster structure? What does the disagreement tell you?
- Do your hierarchical clusters and K-Means clusters tell the same business story?

Day 4 : DBSCAN

Time estimate: 2–3 hours Goal: Run DBSCAN, investigate the noise points, understand what ϵ really means.

What to Understand First: Before running DBSCAN, make sure you understand:

- What is ϵ -neighborhood given a point, what set of points does it define?
- What makes a point a core point, a border point, and a noise point? What are the exact conditions for each?
- What is density reachability? Is it symmetric? What does that mean in practice?
- What is density connectivity? How does it differ from density reachability?
- What does the k-distance plot show? Why does the elbow in this plot give you a principled estimate for ϵ ?
- What are the two parameters of DBSCAN ϵ and min_samples and what does each control geometrically?
- Why does DBSCAN not require you to specify k in advance what it uses instead to determine cluster boundaries?

What to do:

Before running DBSCAN, plot the k-distance plot. This is already scaffolded in your notebook. Look at the elbow. Pick an ϵ from that plot and write down your reasoning.

Run DBSCAN with your chosen ϵ and `min_samples=5`. Look at the output. How many clusters did you get? How many noise points? What percentage of your customers are labeled as noise?

Now pull out those noise customers. Look at their raw RFM values. Who are they? Are they extreme high spenders? Completely inactive customers? One-time buyers? Write a paragraph describing them, not their statistics, their story.

Run at least two more combinations of ϵ and `min_samples`. Build a comparison table: ϵ , `min_samples`, number of clusters, number of noise points, noise percentage. At what point does ϵ become too large? Too small? How do you know?

The key question to sit with today: In DBSCAN, an outlier is not a bad data point it is a point the algorithm refuses to force into a group. Is that refusal informative or is it just a parameter problem? Write your answer.

Day 5: Validation + Comparison

Time estimate: 1.5–2 hours Goal: Compare all three methods honestly. Choose your final segmentation.

What to Understand First: Before computing any metric, make sure you understand what each one actually measures:

- What is the Silhouette Score what do $a(i)$ and $b(i)$ represent, what does a score close to 1 mean versus close to -1?
- What is the Davies-Bouldin Index, what does it penalize, and why does a lower score mean better clustering?
- What is the Calinski-Harabasz Index, what is the ratio it computes, and why does a higher score mean better clustering?
- Why can you not naively compare K-Means and DBSCAN using the same validation metric that makes DBSCAN output structurally different?
- What is the difference between internal validation (no ground truth) and external validation (with ground truth) which you are doing here and what are its limitations?
- What does it mean for clusters to be geometrically coherent but semantically meaningless? Can you construct an example?

What to do:

Compute Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index for K-Means and Hierarchical. For DBSCAN, exclude noise points before computing write a comment explaining exactly why.

Build the comparison table. Look at it. Do the metrics agree across methods? Do they agree with each other within the same method?

Now make a decision: which segmentation are you presenting to the marketing team? Write your justification in two parts: first the metric reasoning, then the business reasoning. If they point to different methods, explain how you resolved the conflict.

The trap to avoid: Do not just pick the method with the highest Silhouette Score. A score of 0.72 with 8 clusters that no marketing team can act on is worse than a score of 0.55 with 4 clear, actionable segments. Metrics are inputs to your decision, not the decision itself.

Day 6: Business Narrative + Failure Log

Time estimate: 1.5–2 hours Goal: Translate your clusters into business language. Complete your failure log.

What to Understand First: No new algorithm concepts today. Before writing, make sure you can answer:

- What does each cluster's RFM profile actually say about that customer's behavior, not the numbers, the behavior?
- What is the difference between a descriptive segment name and an actionable one?
- What would a marketing director need to know to act on your segmentation? What information would they find useless?
- What is a Failure Log for why is documenting what went wrong more valuable than documenting what worked?

What to do:

Give each cluster a name. Not "Cluster 0" a real name. "High-Value Loyalists." "Dormant Bargain Hunters." "New High-Potential." The name should come from the RFM profile, not from aesthetic preference.

Write a 3–4 sentence profile for each cluster. What do their Recency, Frequency, and Monetary values tell you about their behavior? What kind of customer is this?

Write one specific marketing action for each cluster. Not "send them an email." Something specific: "Offer a 15% loyalty discount to customers in this cluster who have not purchased in the last 60 days to reactivate before the holiday season."

Write your executive summary 200 to 300 words. Imagine you are presenting this to a marketing director who has never heard of K-Means. What did you find, why does it matter, and what should the team do with it?

Then complete your Failure Log. Minimum 3 entries. For each: what you expected, what happened, what you learned. Be honest. "I expected K-Means++ to always give better clusters than random init. It did not on my first run. I learned that K-Means++ reduces variance but does not guarantee the globally optimal solution" is a good entry. "It did not work" is not.

If yes: you are ready. If not: go back and fill the gaps.

Session Day:

Time estimate: 2 hours (mandatory)

What to bring:

- Your completed notebook all cells executed, no errors
- At least 3 entries in your Failure Log
- One result that surprised you be ready to describe it in 2 sentences
- One decision you are not fully confident about be ready to defend it anyway
- Your AI conversation log from the AI Discussion Guide

What will happen in the session:

- The session will dissect your experience not teach you from scratch
- Demos will be driven by what students struggled with, not a fixed agenda
- You will be asked to explain your decisions, not just your outputs

What the session is not:

- It is not a lecture you can skip and watch the recording instead
- It is not a place to ask "what is the right answer"
- It is not optional

Quick Reference What to Do When Stuck

Situation

What to do

Cannot build RFM matrix	Re-read the group by logic. Print intermediate outputs at each step. Check if CustomerID is the index.
K-Means gives strange clusters	Check if you scaled. Run with k=2 first simplest case. Print cluster sizes are wildly unequal?
Elbow and Silhouette disagree	Do not panic. Pick both k values, run both, profile both. Choose based on business interpretability.
DBSCAN gives all noise	ϵ is too small. Look at your k-distance plot elbow again. Try doubling ϵ .
DBSCAN gives one giant cluster	ϵ is too large. Halve it.
Dendrogram is unreadable	Use a sample of 200–300 rows. Increase figure size. Use <code>truncate_mode='level'</code> parameter.
Validation metrics conflict	Read what each metric actually measures. They are not measuring the same thing. Document the conflict and your reasoning.
Stuck after 45 minutes on one thing	Post on the forum. Describe exactly where you are stuck, what you tried, and what you expected. Do not post code post reasoning.

Aha! Journal Prompts Fill These In Daily

Write one sentence per prompt each evening. Bring this to the session.

- **Day 1:** How did I think clustering worked before today? What is the first thing that surprised me?
- **Day 2:** What does the instability in K-Means random initialization tell me about the nature of the algorithm?
- **Day 3:** What did the dendrogram show me that a scatter plot could not?
- **Day 4:** Are the DBSCAN noise points in my dataset a failure of the algorithm or information about my customers?
- **Day 5:** If I had to choose one clustering method to use in a production system that runs monthly, which would I choose and why?
- **Day 6:** Where did my intuition fail me this week? What assumption did I make that the data proved wrong?